Atmosphere 3D Weather Application

Technical Explanation & Architecture Blueprint | Generated Documentation

1. Executive Summary

Atmosphere 3D is a modern, condition-reactive weather application inspired by Google Weather. It combines a dynamic 3D WebGL background layer (rendered using React Three Fiber and Three.js) with a minimal, frosted-glass UI built with React, Tailwind CSS, and Framer Motion. The application operates 100% on the client side by leveraging Open-Meteo's free REST APIs without requiring a custom backend server.

2. Technology Stack

Category	Technology	Purpose
Framework	React 18 + Vite	Component architecture, fast HMR dev server
3D Engine	React Three Fiber + Three.js	Declarative WebGL canvas, particle physics
3D Helpers	@react-three/drei	Starfields, floating physics, ambient effects
Animations	Framer Motion	Glass card entrances, hover scale, modal popups
Styling	Tailwind CSS v4 + Vanilla CSS	Glassmorphism, responsive grid, blur filters
Data APIs	Open-Meteo Public APIs	Free Geocoding, Forecast & AQI telemetry

3. Core Architecture & Modules

A. Real-Time 3D WebGL Background Engine

- **WeatherCanvas.jsx**: Master WebGL canvas wrapper with capped Device Pixel Ratio (dpr=[1, 1.25]) for high FPS.
- **CameraParallax.jsx**: Uses mouse cursor tracking and linear interpolation (Lerp) to smoothly tilt the 3D camera.
- **WeatherLighting.jsx**: Coordinates sun/moon directional light positions, sky ambient colors, and lightning flash triggers.
- **Condition Scenes**: Includes *ClearScene* (Sun & corona rays), *NightScene* (Starfield & Moon), *CloudScene* (Drifting clouds), *RainScene* (Instanced rain physics), *SnowScene* (Snowflakes), *FogScene* (Haze volume), and *ErrorScene* (Radar fallback).

B. Client-Side Weather Data Engine (No Backend)

- **openMeteoApi.js**: Directly fetches Open-Meteo Geocoding search suggestions, 7-day daily weather, 24-hour hourly precipitation timelines, AQI telemetry, and GPS reverse geocoding via standard browser fetch().
- **wmoCodes.js**: Converts WMO weather code integers (0–99) into 3D scene configs (sky gradients, cloud counts, rain/snow density).
- **feelsLikeLogic.js**: Evaluates humidity, wind gusts, and UV radiation to generate plain-English apparent temp insights.

C. Google Weather–Style Glass UI & Framer Motion

- **Header.jsx**: City search dropdown with auto-complete, GPS locator, unit switcher (°C / °F), and pinned locations drawer.

- **HeroWeather.jsx**: Large hero temperature, condition title, local time, and feels-like banner.
- **HourlyStrip.jsx**: Horizontal 24-hour carousel with rain probability timeline graph bars.
- **DailyForecast.jsx**: 7-day forecast list featuring min/max temperature range bars.
- **WeatherMetrics.jsx**: Grid of frosted glass chips (AQI rating scale, Wind speed & gusts, UV Index, Humidity, Pressure, Visibility).
- **Framer Motion Animations**: Cards feature staggered entrance sequences, hover scale-up floating effects (whileHover={{ scale: 1.05, y: -4 }}), pulsing background glow auras, and modal popups.

4. Performance Optimizations (Zero Lag)

1. **Device Pixel Ratio Capping**: Set dpr={[1, 1.25]} on Canvas to prevent 4K monitors from rendering 4x–9x excess pixels.
2. **Low-Poly Cloud Geometry**: Simplified cloud sphere segment counts (from 16x16 to low-poly 8x8) and capped maximum clouds to 12.
3. **Particle Density Reduction**: Reduced rain particles to 450 (and snow to 300) while slightly scaling point sizes—saving 80% CPU array updates.
4. **Shadow Map Removal**: Disabled expensive shadow depth map buffer rendering on directional solar lights.
5. **CSS GPU Acceleration**: Reduced CSS backdrop blur to 10px and added will-change: transform to prevent compositor stutter.

5. Project File Structure

```
weather-app/
■■■ src/
■   ■■■ components/
■   ■   ■■■ 3d/              # R3F WebGL Canvas, Lighting & Weather Scenes
■   ■   ■   ■■■ scenes/       # Clear, Night, Cloud, Rain, Snow, Fog, Error
■   ■   ■   ■■■ WeatherCanvas.jsx
■   ■   ■   ■■■ WeatherLighting.jsx
■   ■   ■   ■■■ CameraParallax.jsx
■   ■   ■■■ ui/              # Glassmorphic UI Components & Modals
■   ■       ■■■ Header.jsx, HeroWeather.jsx, HourlyStrip.jsx
■   ■       ■■■ DailyForecast.jsx, WeatherMetrics.jsx, WeatherAlerts.jsx
■   ■       ■■■ SavedLocationsDrawer.jsx, ShareModal.jsx
■   ■■■ context/
■   ■   ■■■ WeatherContext.jsx# Global State, Units (°C/°F), Saved Cities
■   ■■■ services/
■   ■   ■■■ openMeteoApi.js   # Open-Meteo REST API Services
■   ■   ■■■ wmoCodes.js       # WMO Weather Code Map Engine
■   ■■■ utils/
■   ■   ■■■ feelsLikeLogic.js # Rule-based Feels Like Reasoning
■   ■■■ App.jsx
■   ■■■ index.css            # Glassmorphism utilities & Tailwind imports
■■■ package.json
```